

Unknown Factors and Related Division Facts

Answer Key

Grade 2–3 — Foundations of Division

Quick Answers

1. 6	4. 9	7. 7	10. 4
2. 7	5. 6	8. 9	
3. 8	6. 8	9. 48	

Each solution names the unknown-factor equation first, then the multiplication fact that unlocks it. Accept a matching division sentence as equivalent reasoning.

1. Maya’s stickers: 4 equal rows, 24 stickers in all. How many in each row?

Solution: The rows are the known factor, so the equation is $4 \times ? = 24$. Count up by fours: 4, 8, 12, 16, 20, 24 — that is six fours. The related division fact $24 \div 4$ says the same thing. Each row has 6 stickers.

2. 28 seeds planted in equal rows of 4. How many rows?

Solution: Here the row size is the known factor, so the unknown factor counts the rows: $? \times 4 = 28$. From the four-times table, $7 \times 4 = 28$, so the missing factor is 7. The picture confirms it: seven rows of four dots. Number of rows: 7.

3. Find the unknown factor: $7 \times ? = 56$.

Solution: Ask “seven times what makes 56?” Skip-count by sevens: 7, 14, 21, 28, 35, 42, 49, 56 — the eighth number. Check with the related division fact: $56 \div 7 = 8$. The unknown factor is 8.

4. Nina shares 45 stickers equally among 5 friends.

Solution: “5 equal groups, 45 in all” is $5 \times ? = 45$. The five-times table gives $5 \times 9 = 45$, and the related division fact $45 \div 5 = 9$ agrees. Each friend gets 9 stickers.

5. Sam knows $6 \times 8 = 48$. Use it to find $48 \div 8$.

Solution: A division question asks for a missing factor: $48 \div 8$ means $8 \times ? = 48$. Sam’s fact $6 \times 8 = 48$ already supplies that missing factor, because multiplication can be done in either order. So no new work is needed: $48 \div 8 =$ 6.

6. Find the unknown factor: $? \times 9 = 72$.

Solution: The unknown factor is in front, but the question is the same: “what times 9 makes 72?” From the nine-times table, $8 \times 9 = 72$. Check by dividing: $72 \div 9 = 8$. The unknown factor is 8.

7. A tray holds 42 muffin cups in equal rows of 6. How many rows?

Solution: Each row is a group of 6, and the total is 42, so $? \times 6 = 42$. Count by sixes: 6, 12, 18, 24, 30, 36, 42 — seven sixes. Related division fact: $42 \div 6 = 7$. The tray has 7 rows.

8. 6 tables, the same number of chairs at each, 54 chairs in all.

Solution: Six equal groups with a total of 54 gives $6 \times ? = 54$. Nine sixes make 54, since $6 \times 9 = 54$; the related division fact is $54 \div 6 = 9$. Each table has 9 chairs.

9. A mystery number times 7 equals 42. What is that mystery number times 8?

Solution: Step 1 — find the mystery number with an unknown-factor equation: $7 \times ? = 42$, and since $7 \times 6 = 42$, the mystery number is 6. Step 2 — multiply that number by 8: $6 \times 8 = 48$. The answer is 48. (Watch the trap: the question does not ask for the mystery number itself, which is 6.)

10. 48 rolls, 6 rolls per bag, 2 bags per box. How many boxes?

Solution: Step 1 — bags. Each bag is a group of 6 rolls out of 48, so $6 \times ? = 48$, and $6 \times 8 = 48$ gives 8 bags. Step 2 — boxes. Each box holds 2 bags out of those 8 bags, so $2 \times ? = 8$, and $2 \times 4 = 8$ gives 4 boxes. The baker fills 4 boxes.